Image Colorization using Dense Prediction Transformers

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***Abstract*—Traditional image colorization often relies on Convolutional Neural Networks (CNNs), which frequently suffer from color bleeding and inconsistent global semantics due to limited local receptive fields. To address this, we implemented a Dense Prediction Transformer (DPT) from scratch to map grayscale lightness to the LAB color space by leveraging the global self-attention mechanisms of a Vision Transformer (ViT). Our model successfully learned realistic color distributions on the COCO 2017 dataset, significantly outperforming a naive grayscale baseline and demonstrating the viability of transformers for dense pixel-level generative tasks.**

I. Introduction

Image colorization is the highly ill-posed problem of hallucinating plausible color information for a given grayscale input. It has widespread applications in restoring historical photography, enhancing medical imaging, and automating creative workflows. Historically, this domain has been dominated by CNN architectures such as U-Net. However, because standard convolutions operate on localized patches, CNNs often struggle to maintain color consistency across large, disconnected regions of an image (e.g., ensuring a patch of sky in the top-left matches the sky in the top-right).

Our approach addresses these limitations by introducing a Vision Transformer (ViT) backbone. Because ViTs utilize multi-head self-attention, they intrinsically capture long-range dependencies across the entire image space. While transformers were originally designed for image classification [1], frameworks like the Dense Prediction Transformer [2] have adapted them for pixel-to-pixel tasks. By building a custom hierarchical decoder, this project demonstrates an approach to colorization that prioritizes global context over local textures.

II. Teaser Figure

Fig. 1. The proposed DPT architecture processing a grayscale input (L channel) to predict missing color semantics (ab channels), reconstructing a fully colorized RGB output.

III. Approach

A. System Architecture and Implementation

The system was implemented entirely in PyTorch. The architecture follows an encoder-decoder structure:

- **Encoder:** We utilized the official `vit_b_16` pretrained on ImageNet from `torchvision.models` to act as our global feature extractor.
- **Reassemble Blocks:** Transformer tokens from layers 3, 6, 9, and 12 were extracted and projected back into spatial feature maps at various resolutions using custom `ConvTranspose2d` layers.
- **Feature Fusion Blocks:** We implemented Residual Convolution Units to progressively upsample and fuse the feature maps, recovering fine spatial details.
- **Output Head:** The final stage utilizes sequential 3x3 and 1x1 convolutions to output exactly 2 channels. A $\tanh$ activation normalizes the output between -1 and 1.

B. Design Choices and Obstacles

Rather than predicting standard RGB channels, the network operates strictly in the LAB color space. The L (Lightness) channel acts as the input, while the model is tasked only with predicting the 'a' and 'b' color channels. This design choice prevents the network from wasting computational resources reconstructing structural details that already exist in the grayscale image.

The primary obstacle encountered was hardware and time constraints; Kaggle GPU sessions are strictly limited to 12 hours. Training on the full 118,000-image COCO dataset would require roughly weeks. To address this, We mitigated overfitting by initially freezing the ViT backbone to train the random decoder weights, and subsequently unfreezing the backbone midway through training to allow for fine-tuning without catastrophic forgetting.

IV. Experiments and Results

A. Experimental Setup

The model was trained on 118k images from the COCO 2017 dataset, utilizing an 80/20 train/test split. Training was conducted on an NVIDIA Tesla T4 GPU for 5 epochs with a batch size of 8. The base learning rate was set to 2×10^{-4} using the Adam optimizer.

Final Project for CSE 429: Computer Vision and Pattern Recognition. Instructor: Dr. Ahmed Gomaa.

B. Evaluation Metrics and Baselines

We evaluated the model using L1 Loss (Mean Absolute Error). L1 was specifically chosen over Mean Squared Error (MSE) because MSE penalizes variance heavily, often causing generative models to average their predictions into safe, muddy, desaturated colors. L1 encourages more vibrant and distinct color choices.

Our baseline for evaluation is a naive "do nothing" approach (predicting $a = 0, b = 0$ for all pixels, resulting in the original grayscale image).

C. Parameter Analysis and Trends

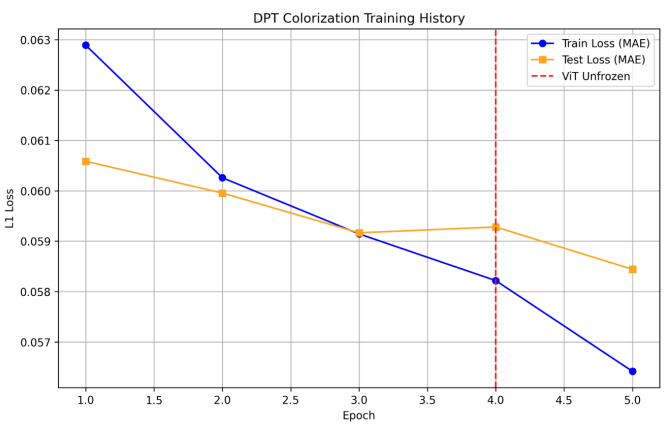

Fig. 2. L1 Loss (MAE) over 5 epochs. The red dashed line indicates the unfreezing of the ViT backbone.

As illustrated in Fig. 2, the network learned rapidly during the first three epochs while the backbone was frozen. At Epoch 4, the backbone was unfrozen and the learning rate was dropped to 1×10^{-5}. This sudden introduction of gradients to millions of parameters caused a temporary shock, reflecting as a slight plateau in the test loss. However, by Epoch 5, the model stabilized and the test loss reached its absolute minimum, proving that unfreezing allowed the ViT to fine-tune its feature extraction specifically for color semantics.

V. Qualitative Results

Fig. 3. Success Case: The model accurately identifies and colorizes natural elements such as sky and foliage based on global context.

Visual outputs confirm the numerical results. As seen in Fig. 3, the model successfully understands global semantics, correctly distributing blue and green hues to their respective environmental contexts.

Fig. 4. Failure Case: The model struggles with artificially colored objects lacking deterministic grayscale lightness cues.

However, the model exhibits expected failure cases (Fig. 4) when dealing with man-made objects (e.g., vehicles, clothing) where the grayscale lightness alone does not provide a deterministic cue for the true color, occasionally defaulting to sepia or gray tones.

VI. Conclusion and Future Work

This project successfully demonstrated that a Dense Prediction Transformer can effectively map grayscale lightness to the LAB color space by leveraging global self-attention. By building the Reassemble and Fusion blocks from scratch, we overcame the local limitations of traditional CNNs.

Future work includes scaling the compute to train on the full 118,000-image COCO dataset for extended epochs to achieve state-of-the-art realism. Furthermore, integrating a Generative Adversarial Network (GAN) discriminator could heavily penalize dull colors, resulting in significantly more vibrant predictions.

References

[1] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *International Conference on Learning Representations (ICLR)*, 2021.

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[3] Y. Zhao et al., "Hint-Based Image Colorization Based on Hierarchical Vision Transformer," *Sensors*, vol. 22, no. 18, 2022.